



A novel method for tracking urban liveability over time: Case study from Melbourne, Australia (2016–2021)

Manoj Chandrabose^{a,b,*}, Lucy Gunn^a, Alan Both^a, Takemi Sugiyama^c, Melanie Davern^a

^a Centre for Urban Research, RMIT University, Melbourne, VIC, Australia

^b School of Health Sciences, Swinburne University of Technology, Melbourne, VIC, Australia

^c School of Social Sciences, Media, Film and Education, Swinburne University of Technology, Melbourne, VIC, Australia

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ABSTRACT

Existing liveability indices are limited as they cannot be used to accurately monitor changes in liveability of a place over time. This study introduces a new method, the Adjusted Mazziotta-Pareto Index (AMPI) approach, to calculate the liveability index and accurately measure its changes. This liveability index was calculated for 403 suburbs in Melbourne, Australia, in 2016 and 2021. It consisted of walkability components, social infrastructure, regular public transport, public open space, local employment opportunities, and housing affordability as the constituent indicators. By employing the AMPI approach, these indicators were normalised into a time-independent scale and aggregated by incorporating a penalty for any imbalances among them. This approach provided a balanced assessment of liveability, which is comparable across multiple time points. Our analysis of within-suburb changes in the liveability index found that, while liveability in most suburbs remained stable over a five-year period, some outer suburbs in Melbourne exhibited considerable changes, primarily driven by changes in access to regular public transport and daily living destinations. Understanding changes in liveability has the potential to advance both research and policy practice on improving liveability. Future research could apply the AMPI-based liveability index to investigate to what extent neighbourhood liveability changes over time can influence residents' health and wellbeing trajectories. Practitioners and policy makers can make evidence-informed decisions in identifying areas where investments are needed to improve liveability.

1. Background

Governments worldwide are increasingly interested in improving the 'liveability' of cities (Asian Development Bank, 2019; Hill et al., 2021). In simpler terms, a liveable city should have communities that promote the health, wellbeing, and quality of life of its residents (Wagner & Caves, 2012). However, in order to develop actionable strategies for improving liveability of local communities, a clearer operational definition is required.

A recent comprehensive review of liveability indicators reported that this concept is generally defined as a multi-dimensional phenomenon typically encompassing environmental, social, and economic dimensions (Khorrami et al., 2021). The term liveability has been popularised through global liveability ranking indices developed by the Economist Intelligence Unit (EIU), which assesses cities on a broad scale, primarily for expatriate executive remuneration (Giap et al., 2014).

Several other similar definitions also exist, such as those proposed by Mercer's Quality of Living Index (Marsal-Llacuna et al., 2015) and the Organisation for Economic Co-operation and Development's (OECD) Better Life Index (OECD, 2024). Additionally, in research, the concept of liveability has been operationalised using different types of indicators and various methodological approaches (Benita et al., 2021; Chen et al., 2023; Ghasemi et al., 2018; Xiao et al., 2022). Although several liveability indicators exist in the literature, the selection of a specific definition depends on the purpose and context of the investigation.

A decade ago, Australian researchers developed a comprehensive definition of 'liveability', through extensive reviews of both international academic and policy literature, as well as incorporating feedback from stakeholders at all levels of government and relevant private sectors (Badland et al., 2014; Lowe et al., 2015). *Liveable communities* were defined as places that are safe, attractive, socially cohesive and inclusive, environmentally sustainable, offer diverse and affordable housing

* Corresponding author at: Centre for Urban Research, 124 La Trobe St, Building 8, Level 11, Melbourne City campus, RMIT University, GPO Box 2476, Melbourne VIC 3001 Australia.

E-mail address: manoj.chandrabose@rmit.edu.au (M. Chandrabose).

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options with convenient access to active travel and public transportation infrastructures that are well connected to employment, education, local shops, health care, community services, public open spaces, leisure and cultural opportunities (Badland et al., 2014; Lowe et al., 2015). Importantly, this definition well aligns with the Social Determinants of Health Framework (World Health Organization, 2010).

Building on this definition, Higgs et al. (2019) developed a composite urban liveability index incorporating several environmental, social, and economic indicator domains. This composite index included indicators such as the walkability index, social infrastructure access, public transport access, public open space access, local employment opportunities, and housing affordability. The primary purpose of developing this index was to map the spatial distribution of liveability across local areas within cities as well as to compare the overall liveability between cities. Since its creation, this index has been extensively used in both research and practice. For instance, it was used to highlight spatial and socio-economic inequalities in liveability within Australian capital and regional cities (Giles-Corti et al., 2022). It was also used to examine cross-sectional associations of neighbourhood liveability with residents' travel behaviours (Higgs et al., 2019) and cardio-metabolic health outcomes (Higgs et al., 2021). Importantly, the Australian Urban Observatory (AUO, auo.org.au) has developed a nationwide digital platform of liveability (using this index) for major Australian cities. Local governments, community organisations, urban and transport planners, and policymakers increasingly rely on this platform for evidence-based decision-making. The impacts and policy applications of the AUO, particularly its benefits for urban planning and public health research and practice, have been previously described in detail by Davern et al. (2023).

Despite its extensive applications, the current liveability index has a key limitation: it cannot be used to accurately track changes in the liveability of a place over time. This is because it is calculated as a measure relative to a specific point in time, rather than as a measure that can be meaningfully compared across multiple time points (Higgs et al., 2019). However, assessing changes in liveability over time is important for several reasons. Cities are constantly evolving due to factors such as population growth (Australian Bureau of Statistics, 2025) and infrastructure development (Infrastructure Australia, 2024). As a result, liveability may increase or decrease due to changes in one or more of its constituent indicators, often influenced by planning policies such as residential development, infrastructure upgrades, provision of amenities or creating new employment opportunities (Infrastructure Australia, 2018). For instance, well-planned higher-density housing developments can enhance liveability by increasing access to amenities, improving walkability, and fostering social connection (Foster et al., 2023). Therefore, accurately measuring these changes can help to determine whether cities are improving their liveability conditions over time. Moreover, monitoring liveability over time can help to track the progress toward achieving the United Nations' Sustainable Development Goal 11: making cities inclusive, safe, resilient, and sustainable. Similarly, previous Australian research has reported health benefits associated with improvements to liveability components such as access to green space (Carver et al., 2024). Therefore, developing a liveability index suitable for longitudinal analysis may help to explore the relationships between changes in neighbourhood liveability over time and health-related outcomes, including physical health (e.g., cardiometabolic risk factors) (Chandrabose et al., 2019) and mental health (e.g., psychological wellbeing) (Carver et al., 2024), as well as health behaviours such as walking (Bandara et al., 2025). It is also important to note that although there are a few other indices such as the EIU's Global Liveability Ranking, Mercer's Quality of Life Index, and the OECD Better Life Index used in practice to track liveability over time, they do not explicitly describe the methodologies used to track changes (Giap et al., 2014; Marsal-Llacuna et al., 2015; OECD, 2024). Furthermore, the OECD Handbook on Constructing Composite Indicators, despite its widespread use in guiding the development of composite indices, also

does not provide a specific method for measuring changes in a composite index over time (European Union - Joint Research Centre, 2008).

Khorrami et al. (2021) offer a comprehensive overview of liveability score calculation methods; however, their review does not give attention to approaches suitable for longitudinal assessment of liveability. As such, there is a clear need for a new approach to calculating a liveability index that enables meaningful comparisons of the liveability of places across multiple time points.

The following sections outline the operational foundations of the previous liveability index developed by Higgs et al. (2019) and detail the additional steps needed to enable longitudinal assessment of liveability. Specifically, we explain the Mazziotta-Pareto Index (MPI) approach (Mazziotta & Pareto, 2016) as the basis for the previous index and introduce the Adjusted Mazziotta-Pareto Index (AMPI) approach (Mazziotta & Pareto, 2017) as an improved method for longitudinal assessment.

1.1. The mazziotta-pareto index approach

The development of the previous liveability index involved several steps: selecting appropriate indicators, ensuring data availability, defining suitable spatial units, normalising the indicators to make them comparable on a common scale, and aggregating those normalised indicators to compute a single index (Higgs et al., 2019). The last two steps are particularly relevant to the focus of the current study. First, to normalise the indicators, they applied the commonly-used z-score approach, which transforms the raw values (by subtracting the mean and dividing by the standard deviation) into unitless values with a mean of 0 and a standard deviation of 1. To aggregate the normalised scores into a composite index, they employed the MPI approach (Mazziotta & Pareto, 2016). A common aggregation method for computing a composite index is summing or averaging the normalised scores (European Union - Joint Research Centre, 2008). However, this method has often been criticised for its compensatory nature, meaning that low values in one indicator can be offset by high values in another (Mazziotta & Pareto, 2013). For example, an area with low walkability can result in a higher level of liveability if that area has a high score for housing affordability. However, low walkability may not be fully compensated by affordable housing. Considering that these liveability indicators represent distinct qualities that may not be interchangeable, the MPI approach, which incorporates a penalty for any imbalances among the constituent indicators [technical details to follow in the Methods sections] (Mazziotta & Pareto, 2016), was used in the development of the original liveability index. This approach would discourage high scores in some indicators from compensating for low scores in others, thus allowing for a more balanced assessment of the target phenomenon.

1.2. The adjusted mazziotta-pareto index approach: introduction of time-independent scales

The main limitation of the original liveability index lies in its use of the z-score method for normalisation, which uses the mean and standard deviation of the indicators at a specific time point to transform the raw values into normalised values. However, if the mean and standard deviation values of an indicator change over time, the normalised values of that indicator for a specific spatial unit can be different at two different time points even if there is no actual change in the indicator's raw value for that unit. This makes the z-score technique unsuitable for normalising the indicators if the purpose of the investigation is to accurately assess changes in liveability over time. To address this issue, the AMPI approach has been introduced as an improvement to the original MPI approach. The AMPI approach applies a time-independent rescaling method for normalising the indicators (detailed in the Methods section), which eliminates artificial shifts in normalised value that can be caused by changes in indicator distributions, thus allowing for valid comparisons across multiple time points (Mazziotta & Pareto, 2017). This

adjustment is particularly important for longitudinal analyses, where the statistical distributions of liveability indicators often change between waves, a consideration typically unnecessary in the development of the liveability index for cross-sectional applications (Higgs et al., 2019). It should be also noted that similar to the MPI, the AMPI approach also includes a penalty for spatial units with imbalanced indicators. Therefore, the application of the AMPI approach to calculate a liveability index can enable meaningful comparisons of liveability across different time points more accurately.

The AMPI approach has been applied to assess changes in multidimensional phenomena in different contexts. For example, it has been used to analyse changes in the quality-of-life index in Italy (Viccaro et al., 2021) and the social vulnerability index in Portugal (Santos et al., 2022). Given the lack of research investigating changes in liveability over time, this paper presents a case study in which the AMPI approach was used to calculate the liveability index at two time points (2016 and 2021) for local areas in Melbourne, Australia. By calculating a liveability index using this new approach, we aimed to understand where and to what extent liveability has changed. In addition, we identify the places with the greatest increases and decreases in overall liveability and examine the key indicators driving these changes. Through this investigation, we aim to demonstrate that the AMPI-based liveability index has potential to be applied as a practical tool by urban planning practitioners to not only monitor the overall liveability of local areas, but also to give insight into the indicators that have changed the most.

2. Methods

2.1. Case study area: greater melbourne

The Greater Melbourne region (referred to here as Melbourne), shown in Fig. 1, is the case study area for this investigation. This region is defined according to the Greater Capital City Statistical Areas classification by the Australian Bureau of Statistics (Australian Bureau of Statistics, 2016). Melbourne is the second most populous city in Australia, with a population of 4.9 million in 2021, increased from 4.5 million in 2016 (Australian Bureau of Statistics, 2024). Melbourne is the fastest-growing capital city in Australia, and it is projected to become the country's largest city by 2032 with a population of 6.2 million (Centre for Population, 2023).

We assessed liveability in 2016 and 2021, the two most recent census years. The decision to use only these two time points was due to the unavailability of the required spatial data for some of the indicators for earlier census years. The spatial unit of analysis for this investigation is a 'suburb' (Australian Bureau of Statistics, 2021b). In 2016, Melbourne had 430 suburbs, which increased to 445 in 2021 due to administrative revisions. To ensure the same spatial units are used in our analysis over time, we used geographical correspondence data to identify suburbs that were spatially consistent across both census years (Australian Bureau of Statistics, 2021a). In total, 403 suburbs were included in our analysis, which are shown in grey in Fig. 1.

2.2. Indicators of liveability

Higgs et al. (2019) used the following six indicators to calculate the



Fig. 1. Map of the Greater Melbourne region, with the suburbs included for the analysis shaded in grey.

urban liveability index: walkability index (a composite index of dwelling density, street connectivity, and access to daily living destinations), social infrastructure access, regular public transport access, public open space access, local employment opportunities, and housing affordability stress. We used the same indicators to calculate a liveability index in our current study. However, instead of the composite walkability index, we included its three components as separate indicators [note that the use of these three indicators to represent walkability has been validated in the Australian context (Mavoa et al., 2018)]. This decision was made because the walkability index is another composite index typically calculated using the z-score normalisation of its indicators. To avoid complications of applying the AMPI approach twice (i.e., once for the walkability index and once for the overall liveability index), we chose to include the individual walkability components separately. Data for these indicators were obtained from the AUO. Detailed information (meta-data) about the data sources and calculation methods are available on the AUO website (<https://auo.org.au/measure/>) and are summarised in Table 1. To align with census years, we selected 2016 and 2021 as the two time points for longitudinal comparison. However, for certain indicators, particularly those related to points of interest such as destinations and social infrastructure, we used OpenStreetMap (OSM) data sourced in 2018, as noted in Table 1. This dataset includes features present prior to 2018, making it a reasonably close proxy for 2016. However, it is more reasonable to consider this comparison of liveability as reflecting a 3-to-5 years change, rather than a strict a five-year change. Prior research has demonstrated the usefulness of OSM data to effectively model urban change (Zhang & Pfoser, 2019).

2.3. Calculating the liveability index using the AMPI approach

Two calculation steps are required to compute a composite liveability index using the AMPI approach: (i) normalising each indicator to convert them into a time-independent common scale, and (ii) aggregating those normalised indicators to compute a single liveability index score (Mazziotta & Pareto, 2017). These steps are described below.

2.3.1. Indicator normalisation

For the current analysis, we have 8 indicators for 403 suburbs at two time points. In the following formulas, x_{ijt} denotes the value of the indicator j for suburb i at time point t . The normalisation process of an indicator depends on its polarity, which can be either positive or negative. An indicator is known to have positive polarity if higher values of it contribute to higher levels of the composite index, and negative polarity if higher values contribute to lower levels of the index (Mazziotta & Pareto, 2017). In the current study, referring to Table 1, dwelling density, daily living destinations access, intersection density, social infrastructure access, public transport access, and local employment opportunities were measured such that higher values of them contribute to higher levels of liveability, and therefore have positive polarity. Conversely, public open space access (measured as distance to closest public open space, thus higher values of this measure contribute to lower levels of liveability) and housing affordability stress (measured as the percentage of households experiencing affordability stress, thus higher values contribute to lower levels of liveability) both have negative polarity (Higgs et al., 2019). In the AMPI approach, the min-max transformation method is used to rescale the raw values (x_{ijt}) into normalised values (r_{ijt}) using the following formula.

$$r_{ijt} = \begin{cases} \left(\frac{x_{ijt} - \text{Min}_{x_j}}{\text{Max}_{x_j} - \text{Min}_{x_j}} \right) \times 60 + 70; & \text{if indicator } j \text{ has positive polarity} \\ \left(\frac{\text{Max}_{x_j} - x_{ijt}}{\text{Max}_{x_j} - \text{Min}_{x_j}} \right) \times 60 + 70; & \text{if indicator } j \text{ has negative polarity} \end{cases}$$

Note that for indicators with negative polarity, the normalisation

Table 1
Details of liveability indicators.

Liveability indicators	Description	Data source	Polarity *
Dwelling density (walkability component)	Average of total number of dwellings within 1600 m road network buffer from all residential addresses in the suburb	ABS 2016 & 2021 census data	Positive
Daily living destination access (walkability component)	Average of a binary indicator representing the presence of three key destinations (supermarkets, public transport stops, convenience stores) within 1600 m road network buffer from all residential addresses in the suburb	OSM 2018 & 2021 points of interest data, GTFS 2018 & 2021 data, and publicly available supermarket location data	Positive
Intersections density (walkability component)	Average of total number of 3-or-more street intersections within 1600 m road network buffer from all residential addresses in the suburb	OSM 2018 & 2021 road network data (only pedestrian-accessible roads were included by excluding freeways)	Positive
Social infrastructure access	Average of an indicator representing the presence of 16 difference social services relevant to health, education, community, sports, and culture in the suburb. Different buffer sizes were used for different services, as described by Higgs et al. (2019).	A range of publicly available data sources for 2018 & 2021 were used as listed on the AUO website.	Positive
Public transport access	Percentage of dwellings in the suburb within 400 m of public transport with a regular 30-minute weekday service (7am and 7pm)	GTFS 2018 & 2021 data	Positive
Public open space access	Average of the distances to closest public open space larger than 1.5 hectares from all residential addresses in the suburb	OSM 2018 & 2021	Negative
Local employment opportunities	Percentage of employed persons in the suburb working in the same local government area	ABS 2016 & 2021 census data	Positive
Housing affordability stress	Percentage of households in the suburb whose incomes were at bottom 40 % but spending more than 30 % of income on housing costs	ABS 2016 & 2021 census data	Negative

Note: ABS – Australian Bureau of Statistics, OSM – Open Street Maps, GTFS - General Transit Feed Specification.

* An indicator has positive polarity if higher values of it contribute to greater liveability and negative polarity if higher values contribute to lower liveability.

process inverts the values so that after normalisation, higher values contribute positively to liveability. The rescaling is designed so that the mean value of each indicator at baseline year (2016 in our study) becomes the reference value of 100 once it is normalised, and the range of normalised values is constrained between 70 and 130 (referred to as ‘goalposts’). In this way, the resulting normalised scores of the indicators are comparable to those calculated by Higgs et al. (2019), who

set the mean at 100 as a ‘natural reference point’ to facilitate easier interpretation: i.e., values higher than 100 indicate above-average levels, and values lower than 100 indicate below-average levels. [Note: Higgs et al. (2019) used z-score normalisation, adjusting the mean to 100 and the standard deviation to 10, so that almost all normalised values would fall within three standard deviations, i.e., between 70 and 130. To ensure comparability, we also set our goalposts to the same].

To achieve this, the recommended approach (Mazziotta & Pareto, 2017) for setting Max_{x_j} and Min_{x_j} is as follows:

$$Min_{x_j} = Ref_{x_j} - \Delta_{x_j}$$

$$Max_{x_j} = Ref_{x_j} + \Delta_{x_j}$$

where, Ref_{x_j} is the baseline mean value of the indicator and Δ_{x_j} is defined as the half of difference between the overall maximum ($OverallMax_{x_j}$) and overall minimum ($OverallMin_{x_j}$) values of the indicators across all time points. The formula for Δ_{x_j} is,

$$\Delta_{x_j} = (OverallMax_{x_j} - OverallMin_{x_j}) / 2$$

Substituting these into the normalisation formula above ensures that when an indicator’s raw value equals the baseline mean, it is converted into a normalised score of 100; a value equal to the overall maximum is converted to 130; and a value equal to the overall minimum is converted to 70.

2.3.2. Indicator aggregation

Once the indicators are normalised, the next step is to compute the composite urban liveability index value for each suburb i at each time point t . This is done using the following aggregation formula:

$$ULI_{it} = M_{it} - S_{it} cv_{it}$$

where,

- ULI_{it} : Urban Liveability Index for the suburb i at time point t
- M_{it} : Mean of normalised indicator values for the suburb i at time point t
- S_{it} : Standard deviation of normalised indicator values for the suburb i at time point t
- cv_{it} : Coefficient variation of normalised indicator values for the suburb i at time point t , calculated as S_{it}/M_{it}

This aggregation formula consists of two parts: a measure of the average and a measure of the amount of imbalance. The purpose of including the second part is to penalise the suburbs that have a greater imbalance among the individual indicator values. This is consistent with the MPI approach used to calculate the original liveability index (Higgs et al., 2019). However, the difference in the current AMPI-based approach is its normalisation process, where the indicator values for each suburb across multiple time points are rescaled into time-independent values. This allows for changes in the liveability index and its indicators in a specific place to be assessed on the same basis over time.

2.4. Analysing the changes in the liveability index

The change in the liveability index for each suburb from 2016 to 2021 was calculated by subtracting the 2016 value from the 2021 value. Descriptive statistics were calculated for both raw and normalised values of the individual indicators, the liveability index in 2016 and 2021, and the changes in the liveability index and its indicators. The spatial distribution of changes in liveability from 2016 to 2021 across suburbs in Melbourne was visualised using a choropleth map with a continuous colour scale. Spatial autocorrelation analysis was conducted

by calculating Global Moran’s I statistic to examine the spatial patterns (i.e., clustered, random or dispersed) of changes in the liveability across suburbs. Moran’s I can range from -1 to $+1$, where positive scores indicate clustering of similar values in neighbouring suburbs and negative scores suggest neighbouring suburbs with dissimilar values (Gaetan & Guyon, 2010). Specifically, a Moran’s I statistic value close to $+1$ (above 0.7) indicates strong spatial clustering of similar values, between 0.3 and 0.6 indicates a moderate level of spatial clustering, and near 0 (below 0.3) suggests weak or no clustering (random pattern). This analysis was conducted using the R package ‘spdep’, with neighbouring suburbs defined as those sharing a boundary and row-standardised weights were applied to account for the differences in the number of neighbouring units between suburbs. We provided supplementary materials showing the spatial distributions of the liveability index for 2016 and 2021, as well as the changes in normalised values for each indicator. One useful aspect of the normalisation method used to calculate the AMPI-based liveability index is that it enables direct comparisons (since they are transformed onto a comparable common scale) of changes in individual indicators over time. In addition, we also identified the suburbs with the greatest increases and decreases in the liveability index (top 2 for each) and used radar plots, which is a graphical display that allows for the relative comparison of multiple variables, to show which specific indicator changes contributed most to changes in the liveability index in those suburbs.

3. Results

Table 2 presents the descriptive statistics (mean and standard deviation) for liveability indicators in 2016, 2021, and their changes between this period. These are provided for both raw values (in the units of

Table 2

Mean (SD) of liveability indicators in 2016, 2021, and their changes over the period ($N = 403$ suburbs).

Indicators	Mean (SD)		
	2016	2021	Change
<i>Raw values</i>			
Dwelling density (counts)	3638 (3695)	4092 (4593)	455 (1158)
Daily living destinations score (out of 3)	1.4 (0.7)	1.4 (0.7)	0.0 (0.2)
Intersections density (counts)	208.5 (151.8)	217.1 (155.2)	8.5 (14.7)
Social infrastructure score (out of 16)	5.2 (3.1)	5.1 (3.0)	−0.1 (0.6)
Public transport access (%)	54.3 (29.5)	43.8 (32.0)	−10.5 (16.5)
Public open space access (m)	731.0 (673.7)	656.9 (383.6)	−74.2 (441.4)
Local employment opportunities (%)	30.5 (12.9)	31.6 (12.4)	1.1 (3.2)
Housing affordability stress (%)	15.7 (4.8)	13.5 (4.2)	−2.2 (2.3)
<i>Normalised values</i>			
Dwelling density	100.0 (5.8)	100.7 (7.3)	0.7 (1.8)
Daily living destinations score	100.0 (14.0)	100.3 (14.1)	0.3 (3.4)
Intersections density	100.0 (9.3)	100.5 (9.5)	0.5 (0.9)
Social infrastructure score	100.0 (13.0)	99.5 (12.6)	−0.5 (2.7)
Public transport access	100.0 (17.7)	93.7 (19.2)	−6.3 (9.9)
Public open space access ^a	100.0 (4.6)	100.5 (2.6)	0.5 (3.0)
Local employment opportunities	100.0 (8.9)	100.8 (8.6)	0.8 (2.2)
Housing affordability ^a	100.0 (6.2)	102.9 (5.4)	2.9 (3.0)
Liveability index	99.1 (6.4)	98.9 (6.8)	−0.2 (1.8)

Note: ‘Raw values’ are in the original units of each indicator; ‘normalised values’ are rescaled scores using the ‘min-max’ transformation, with the reference value of 100 was set as the baseline mean.

The ‘change’ values of the indicators were calculated by subtracting their 2016 values from the 2021 values.

each measure) and normalised values (in unitless, time-independent, comparable scale) of the indicator measures. The mean changes in normalised values for most indicators are minimal, except for public transport access and housing affordability. On average, in the normalised scale, access to public transport decreased by 6.3 units, while housing affordability increased by 2.9 units. The changes in access to regular public transport showed greater variability as well ($SD = 9.9$). The average change in the liveability index was also minimal (mean = -0.2 units); however, there was some variability in the liveability change scores ($SD = 1.8$).

Fig. 2 shows the spatial distribution of within-suburb changes in the liveability index from 2016 to 2021. The liveability change scores across the suburbs ranged from -6.5 to 10.6 . The colour gradient in this map represents increases in green, decreases in red, and no change in yellow. Most suburbs are shaded yellow, indicating little to no change in the liveability index. A few suburbs in the northwest outskirts are shaded greener, indicating notable increases in the liveability index. Several outer suburbs in the west, north, and east are shaded red, indicating

decreases in the liveability index. Moran's I statistic value for changes in the liveability index across these suburbs is 0.38 ($p < 0.001$), suggesting a modest level of spatial clustering.

Supplementary Figure S1 shows the spatial distributions of the liveability index across Melbourne suburbs for 2016 and 2021. Moran's I values for the liveability index were 0.79 ($p < 0.001$) in 2016 and 0.82 ($p < 0.001$) in 2021. Maps showing the spatial distributions of changes in the normalised values of specific indicators are presented in Supplementary Figures S2.1 to S2.8. Since these change scores can range from -60 to $+60$ (i.e., normalised scores of the indicators at one time point lie between 70 and 130), the same colour palette for this range is used for the maps of changes of all indicators to enable comparisons between them. These maps show that dwelling density, intersection density, social infrastructure access, public open space access, and local employment opportunities changed minimally in most suburbs between 2016 and 2021. However, access to daily living destinations, regular public transport, and housing affordability changed considerably in some suburbs. Specifically, access to daily living destinations increased in

Change in Liveability Index (2016-2021)

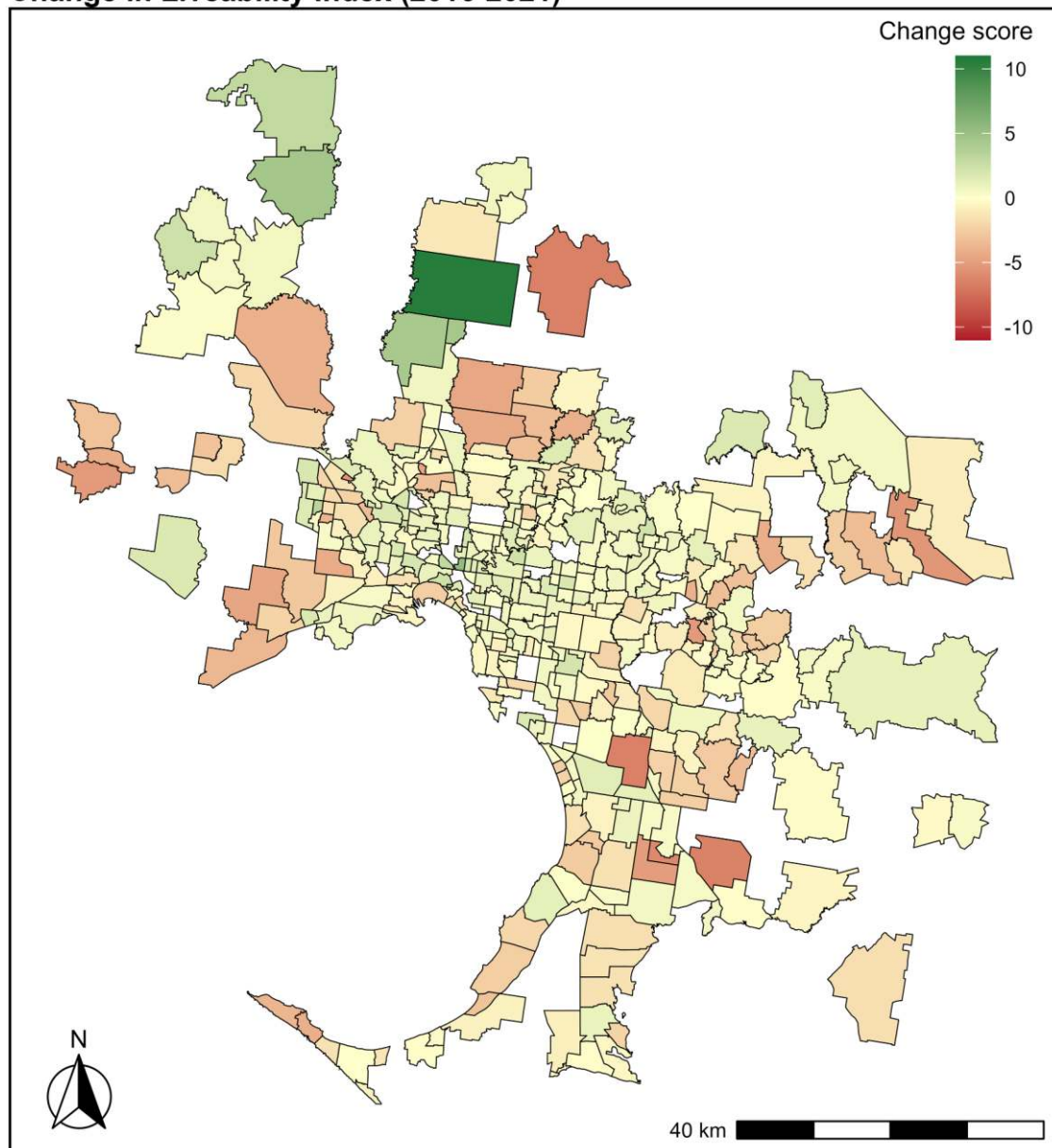


Fig. 2. Spatial distribution of changes in the liveability index across Melbourne suburbs from 2016 to 2021.

some outer suburbs but slightly decreased in others. Access to regular public transport largely declined in most suburbs. Housing affordability slightly improved across most suburbs. The Moran's I statistic values for the indicator change scores ranged from 0.11 (for daily living destinations) to 0.57 (for dwelling destinations).

Fig. 3 shows radar plots illustrating changes in each liveability indicator for four suburbs: the two with the largest increases in the liveability index (Beveridge, +10.6 units, and Carlton, +5.5 units) and the two with the largest decrease in the liveability index (Whittlesea and Dandenong South, both with -6.5 units). The locations of these suburbs are shown in the map in Figure S3. Improvements in public open space access in Beveridge (in the north) and housing affordability in Carlton (near the city centre) appeared to contribute most to their increases in the liveability index. Declines in public transport access were the primary factor reducing the value of the liveability index in both Dandenong South (in the southeast) and Whittlesea (in the north). Interestingly, in Dandenong South, although local employment opportunities increased substantially, larger decreases in public transport access and daily living destination access dominated, resulting in an overall decline in the liveability index.

4. Discussion

Building on earlier research by Higgs et al. (2019), which developed a liveability index suitable only for cross-sectional analysis, the present study introduced a new method (the Adjusted Mazziotta-Pareto Index

approach), to calculate the liveability index that can be used in longitudinal analysis. This method employs a time-independent normalisation technique to rescale the individual indicators of liveability, allowing for a meaningful comparison of the liveability index across multiple time points, thereby providing an accurate assessment of changes in liveability over time. As a case study, we calculated the AMPI-based liveability index for 403 suburbs in the Greater Melbourne region for 2016 and 2021, and analysed the changes in the liveability across these areas. Many suburbs experienced little to no change in liveability. However, there were a few suburbs that exhibited increases in liveability, while some suburbs tended to have decreases in liveability.

Our AMPI-based liveability index is useful for identifying areas with changes in liveability and understanding the contribution of specific indicators to the change. As shown in Figures S2.1 to S2.8, indicators such as dwelling density, intersection density, social infrastructure, public open space, local employment opportunities, and housing affordability changed minimally across most suburbs, particularly in the inner suburbs (Newton et al., 2022). This may reflect the presence of mature infrastructure in these areas, with little policy-driven change during the study period. A few outer suburbs also showed improvements in indicators such as access to public open space (Victoria State Government, 2022) and local employment opportunities (Infrastructure Australia, 2024), reflecting development occurring in outer growth areas. In contrast, most notable declines were seen in access to regular public transport and access to daily living destinations (supermarkets,

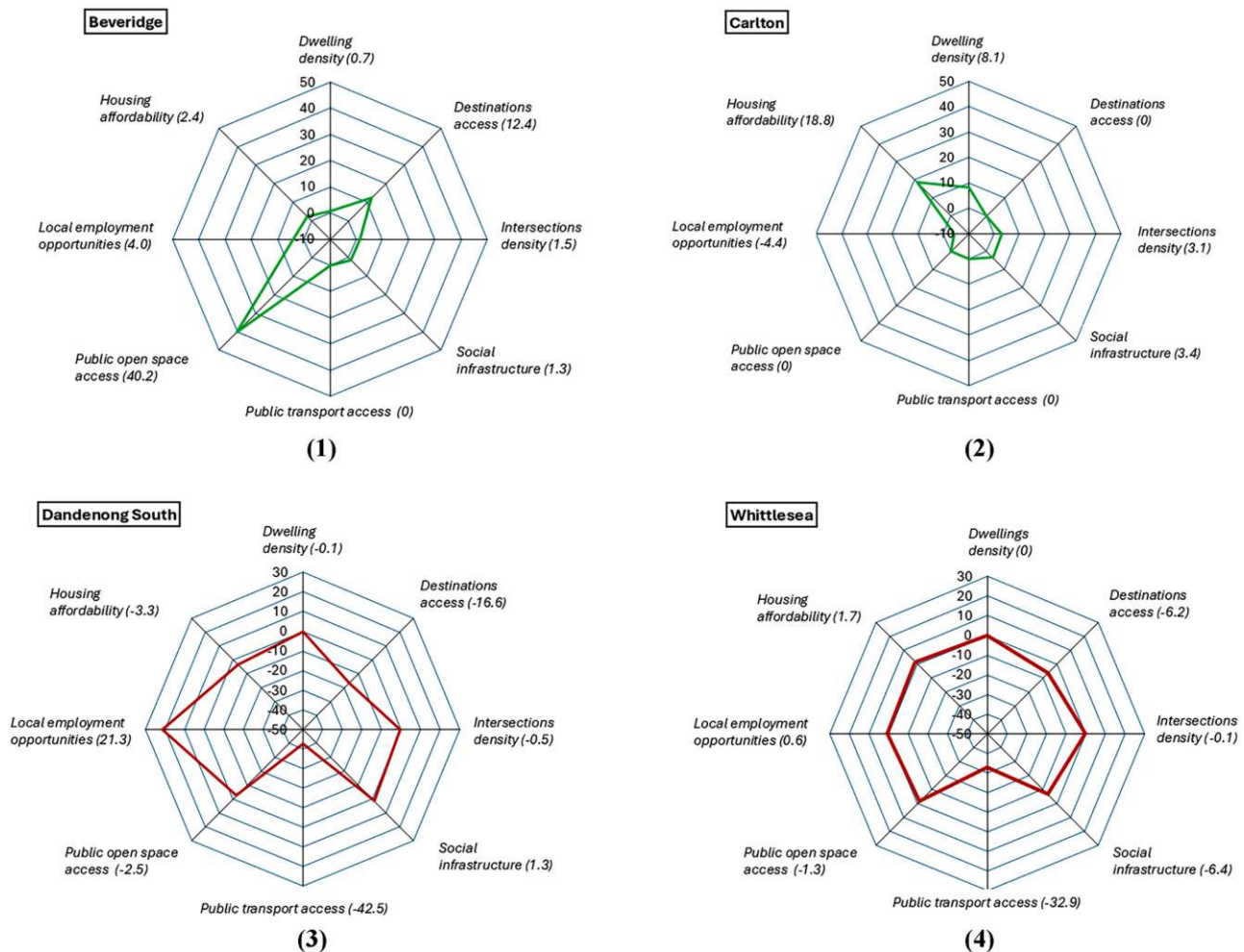


Fig. 3. Radar plots illustrating indicator contributions to changes in liveability index in four suburbs: top 2 increases in liveability (1. Beveridge with 10.6 units increase, 2. Carlton with 5.5 units increase) and top 2 decreases liveability (3. Dandenong South with 6.5 units decrease, 4. Whittlesea with 6.5 units decrease).

public transport stops, and convenience stores). The results suggest that changes in liveability of the suburbs depended more on these two indicators. It should also be noted that declines in these indicators are largely attributable to service restrictions that occurred during the COVID-19 pandemic in 2021 (Stobart & Duckett, 2022). Nonetheless, the findings indicate that some outer suburbs were more adversely affected by the pandemic than inner suburbs, with notable implications for the liveability of those areas. However, these declines may not represent a true reduction in services, as they may have been restored following the pandemic. When data for subsequent years become available, future research could examine whether the observed changes in liveability indicators were temporary disruptions or reflect lasting shifts.

Using radar plots, we identified key indicators contributing to the changes in liveability in four suburbs. Beveridge in the outer northwest had an increase in the liveability index due to improved access to public open spaces and daily living destinations, while Carlton, an inner-city suburb, experienced improvements in housing affordability and dwelling density. Conversely, the greatest decreases in the liveability index were observed in two outer suburbs (Dandenong South and Whittlesea), primarily due to reduced access to regular public transport. While a detailed discussion of the implications of these changes is beyond the scope of the present study, we emphasise that our approach can be useful for understanding how liveability has substantially changed over time in certain areas.

Our analysis showed that liveability changes across Melbourne suburbs between 2016 and 2021 exhibited only a modest level of spatial clustering, with Moran's *I* around 0.4. This indicates that while nearby suburbs experienced somewhat similar changes in the liveability index, the clustering was not particularly strong. It is worth noting that the liveability index itself across these suburbs were strongly clustered, with Moran's *I* close to 0.8, as it was observed in a previous study that used the MPI-based liveability index (Giles-Corti et al., 2022) and replicated in our current analysis of AMPI-based liveability index for both 2016 and 2021 (Supplementary Figure S1). Specifically, inner suburbs surrounding the Melbourne central business district tend to have higher liveability values and outer suburbs typically have lower values. This pattern can be attributed to inner suburbs being generally well-established, high-density areas with greater access to daily living destinations, social infrastructure, public transport, recreational and employment opportunities. In contrast, lower-density outer suburbs have limited access to such amenities. However, it is not surprising that there was no strong spatial clustering in changes in the liveability index scores over the five-years period. This is because planning decisions that aim at improving liveability indicators (e.g., residential developments, building new amenities, providing affordable housing, building new public open spaces, and creating local employment opportunities) are often implemented in a fragmented manner, rather than as a coordinated and integrated approach across neighbouring suburbs.

The new AMPI-based liveability index has the potential to advance both research and practical applications related to liveability. As liveability is well aligned with the current understanding of the social determinants of health (Badland et al., 2014), neighbourhood-level changes in liveability over time could have impacts on residents' health and wellbeing. Longitudinal investigations exploring the relationship between changes in neighbourhood liveability (measured using our new method) and changes in health outcomes could help to generate new knowledge on the links between environments and health. Furthermore, this new liveability index can be utilised to monitor the liveability of Australian cities over time through the AUO platform (Davern et al., 2023), which plans to continue releasing liveability indicator data for Australian local areas on a three-year cycle. Specifically, the previous MPI-based liveability index was used by the AUO to produce scorecards for 21 major cities and their local communities. The new AMPI-based liveability index offers the opportunity to produce liveability scorecards in a temporally consistent manner, enabling accurate

longitudinal comparisons of liveability scores across Australian cities.

Our study has several policy implications. Governments could adopt the AMPI-based liveability monitoring as a framework to assess the performance of local areas and evaluate the impacts of policy interventions. For example, in Melbourne, initiatives such as the 20-Minute Neighbourhood aim to ensure residents can meet most of their daily needs within a 20-minute return walk from home (Victoria State Government, 2019), and the Plan Melbourne 2017–2050 is a long-term strategy designed to guide urban growth in line with these principles (Victoria State Government, 2017). In partnership with the AUO, local governments can use the AMPI-based liveability index and indicators measured across multiple time periods to monitor progress toward these goals, identify areas with limited access to essential amenities and services, and guide where targeted efforts and investments are needed for improvement.

There are some strengths in applying the AMPI approach to measure the liveability index and its change. It is conceptually simple and straightforward to calculate. To apply this approach to cities in other countries, it first requires the calculation of relevant indicators that reflect the key domains of liveability. It depends on the willingness of data custodians to share appropriate datasets for the development of liveability indices. Nevertheless, there are emerging efforts to address these challenges. For example, the *Global Healthy and Sustainable Cities Indicators* platform is working towards creating standardised spatial indicators using open-source data for cities across multiple countries (Higgs et al., 2024). Although these efforts are currently focused on creating indicators for a single time point, in the future they could provide the necessary data across multiple time points. When appropriate indicators at multiple time points are available, the formula presented in the Methods section can be applied to calculate longitudinally consistent measures of liveability for other cities. Statistical tools, such as the R package 'Compind', includes functions for applying this formula. In addition to liveability, this approach can also be applied to calculate other composite indices of environmental attributes such as the walkability index. A recent US study investigated changes in neighbourhood walkability from 2013 to 2020 (Frank & Wali, 2024), but it did not seem to employ a suitable method that enables the comparison between multiple time points. Moreover, while substantial research has explored longitudinal associations between changes in walkability and health outcomes (Chandrabose et al., 2019), limited attention has been given to the development of methods for accurately assessing changes in walkability itself. As defined earlier, in the context of the current study, the walkability index is treated as a component of the broader liveability index, which includes additional indicators beyond those typically used to measure walkability. Nevertheless, since both walkability and liveability are composite indices, future research aiming to examine longitudinal relationships between changes in these composite environmental indices and health outcomes could benefit from adopting the AMPI approach. Furthermore, this new method can also be used to calculate other multi-dimensional composite indices, such as the Better Life Index (OECD, 2024), that are used to track societal progress across countries over time. Another limitation of our study is that our analysis was primarily descriptive, but extending it to explore the factors influencing changes in liveability over time is an important avenue for future research. Additionally, the liveability index used in our study does not include some of the indicators mentioned in the original definition of liveability, such as safety from crime and social cohesion (Lowe et al., 2015). Moreover, in addition to the macro-level built environment features included in our liveability index, micro-level features, such as street tree canopy, accessibility and the quality of footpaths and cycling infrastructure, could also play a role in residents' health and wellbeing. Future research could enhance the liveability index by including additional indicators.

5. Conclusion

Improving the liveability of cities and their local communities is a global policy priority (United Nations, 2024). Achieving this goal requires accurate assessment of changes in liveability of places over time. Our study presents a novel method to assess the liveability index and its changes over time. Using this new approach, we found that the liveability index of most inner suburbs in Melbourne, Australia remained stable from 2016 to 2021. However, considerable changes in the liveability index were observed in some suburbs, particularly in outer suburbs. We also found that changes in two liveability indicators (access to regular public transport options and daily living destinations) were more heterogeneous across suburbs than others (e.g., dwelling density). While our case study observed some changes in liveability index in certain areas, it remains unclear whether such changes have had any impact on residents' health. Empirical research investigating the relationships of neighbourhood liveability with health-related outcomes, including both physical and mental health, is still in its early stages and is predominantly cross-sectional in design (Higgs et al., 2019, 2021). In addition, most existing longitudinal research studies in the built environment and health field have focused on specific components of liveability, such as access to public open space (Carver et al., 2024) or the walkability index (Chandrabose et al., 2019). Future research could expand knowledge in this area by employing longitudinal designs to explore the potential prospective associations between changes in neighbourhood liveability and changes in residents' health outcomes. This can be investigated using existing longitudinal health cohort datasets with several waves of data collection and by calculating the neighbourhood liveability index within participants' neighbourhoods at multiple time points. Analysing the relationships between changes in neighbourhood liveability (resulting from infrastructure improvements or residential relocation) and people's health outcomes could provide better insights into the potential causal links between liveability and health.

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CRediT authorship contribution statement

Manoj Chandrabose: Writing – original draft, Visualization, Supervision, Methodology, Formal analysis, Conceptualization. **Lucy Gunn:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Alan Both:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Takemi Sugiyama:** Writing – review & editing, Methodology, Conceptualization. **Melanie Davern:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

None

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scs.2025.106735.

Data availability

Data will be made available on request.

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